Deep Learning and protein embedding at the Bologna Biocomputing Group

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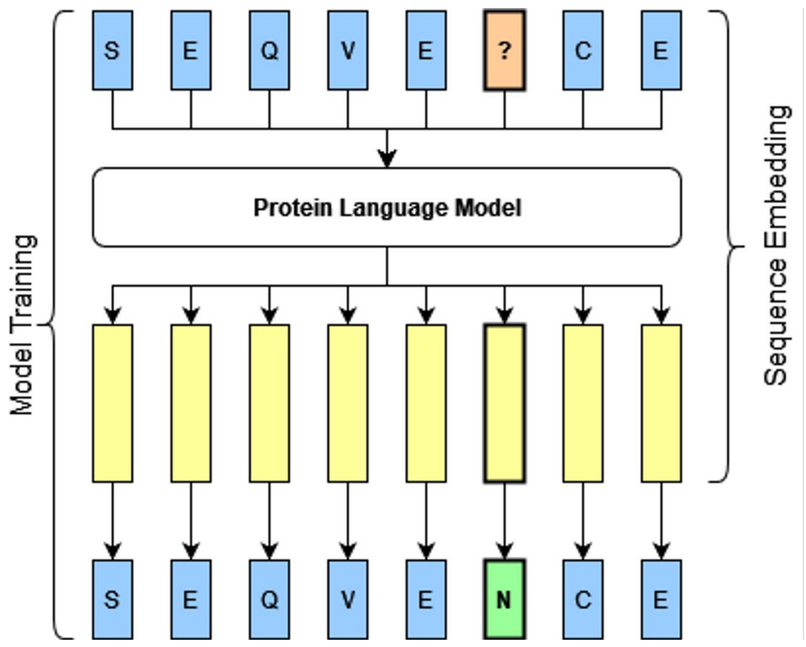
Embeddings from protein language models

Recent tools based on Deep Learning take advantage of protein language models (PLMs, such as ProtT5 and ESM1) for protein encoding. PLMs emerged as a promising framework for protein representation, providing an effective and time-efficient alternative to standard encodings based on evolutionary information. These models are pre-trained in a task-independent way on large databases of protein sequences (e.g., UniRef and/or BFD). Pre-trained models are released to compute meaningful representations of proteins and residues, to be in turn adopted in input for different downstream prediction tasks (a process called fine-tuning).

Here we show two examples of successful applications of embeddings generated by PLMs.

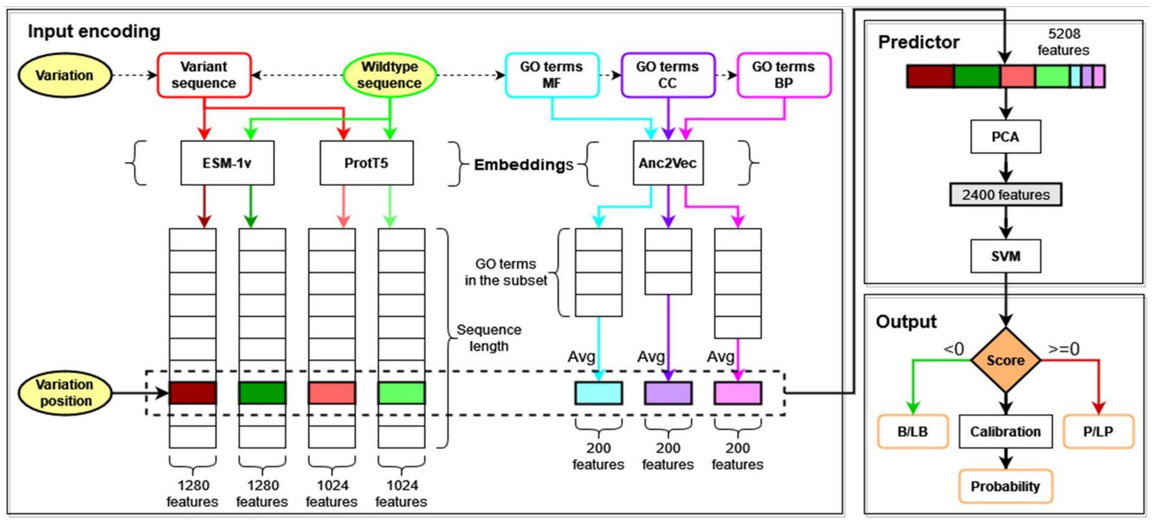


E-SNPs&GO

Prediction of pathogenicity of Single Residue Variations

E-SNPs&GO adopts protein embeddings, together with Gene Ontology terms annotated on the protein, to create a vectorial representation of a Single Residue Variation. A machine learning-based model exploit this encoding to compute a probability for the variation to be disease associated.

E-SNPs&GO achieves state-of-the-art performances when benchmarked on a blind test set. The method is freely available as a web server (link in the QR code above).



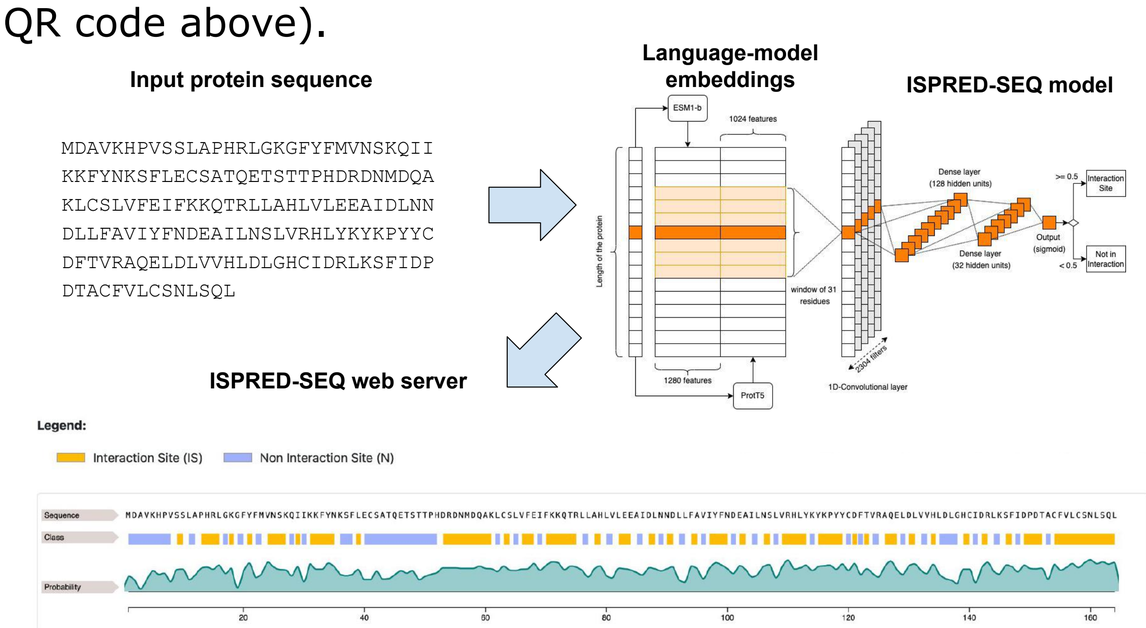
Method	Q_2	Precision	Recall	F1-score	ROC-AUC	MCC
E-SNPs&GO	**0.87**	**0.86**	**0.80**	**0.83**	**0.86**	**0.72**
SNPs&GO	0.80	0.85	0.63	0.72	0.78	0.58
MutPred2.0	0.86	0.79	0.88	0.83	0.86	0.71
PROVEAN	0.78	0.69	0.83	0.75	0.79	0.57
SIFT	0.74	0.63	0.88	0.73	0.77	0.53
PolyPhen-2	0.72	0.60	0.90	0.72	0.75	0.50

ISPRED-SEQ

Prediction of Protein-Protein Interaction SIte

Similarly, ISPRED-SEQ scans a whole protein sequence to distinguish residues that are likely to be involved in Protein-Protein Interactions. Each one is represented with a matrix composed by the concatenated embeddings of neighbouring residues.

ISPRED-SEQ significantly surpasses other state-of-the-art methods when benchmarked on a blind test set. The method is also freely available as a web server (link in the QR code above).



Method	Q_2	Precision	Recall	F1-score	ROC-AUC	MCC
ISPRED-SEQ	**0.86**	**0.47**	**0.47**	**0.47**	**0.82**	**0.39**
PITHIA	0.84	0.41	0.41	0.41	0.78	0.32
DELPHI	0.83	0.37	0.37	0.37	0.74	0.27
PIPENN	0.79	0.39	0.39	0.39	0.73	0.25
SCRIBER	0.82	0.33	0.33	0.33	0.72	0.23
SSWRF	0.81	0.29	0.29	0.29	0.69	0.18

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